Evaporation

Exam code: 0625 & 0972

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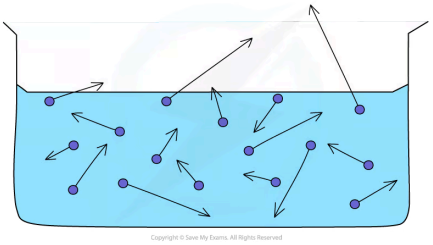
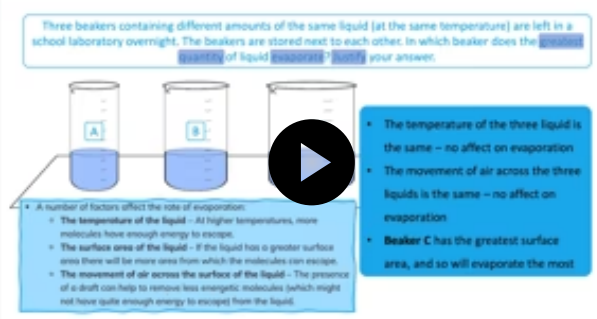
Evaporation

- **Evaporation** is a change in state of a liquid to a gas
- It happens:
  - At **any** temperature
  - Only from the **surface** of a liquid
- The molecules in a liquid have a **range** of energies
  - Some have lots of energy, others have very little
  - Their **average energy** relates to the temperature of the liquid
- Evaporation occurs when **more energetic molecules** moving near the surface of the liquid have enough energy to **escape**
  - The average energy of the liquid is **reduced** when the particles with most energy leave
  - Therefore liquids are **cooled down** by evaporation



Evaporation occurs when more energetic molecules near the surface of a liquid escape



Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Factors affecting evaporation & explaining cooling

Extended tier only

Factors affecting evaporation

- These factors all affect the rate of evaporation

Temperature of the liquid

- Increased temperature increases the **kinetic energy** of the molecules in the liquid
  - Molecules with more energy are more likely to **overcome the intermolecular forces** holding them in the liquid state and escape the surface
  - Therefore **higher temperature** leads to a **higher rate of evaporation**

Surface area of the liquid

- Molecules only **escape** the intermolecular forces of attraction at the **surface** of the liquid
  - Therefore a **larger surface area** leads to a **higher rate of evaporation**

Air movement

- Air movement carries away the water vapour which has just evaporated
  - This replaces the air above the liquid with drier air, which accept water vapour more easily
  - Therefore **increasing air movement** (e.g. wind or a fan) **increases the rate of evaporation**

Evaporation & cooling

- The process of evaporation can be used to cool things down
- As evaporation occurs, the liquid cools
  - This is because the most energetic particles leave, **reducing** average kinetic energy
- Placing an object next to this liquid **cools** the object
  - This is because the cooler liquid **absorbs thermal energy** from the object
  - This process is used in some refrigerators and air conditioning units

Boiling vs evaporation

Extended tier only

- Boiling is **also** a change in state from liquid to gas
  - Boiling happens only at the boiling point of the liquid
  - The change of state happens **throughout** the liquid (seen as bubbles at the bottom of a pan of boiling water, for example)

Table showing the difference between evaporation and boiling

	Evaporation	**Boiling**
Change of state?	Liquid to gas	Liquid to gas
Temperature?	Any temperature between melting and boiling point	Boiling point
Location in liquid?	From the surface	Throughout the whole liquid

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